

HW7 Report: Streamflow Forecasting Workflow

HWR5 564B | Aamir

GitHub Repo: <https://github.com/aamirlc/hw7-streamflow>

Exercise 3 Drawing

Streamflow Forecasting Workflow | Verde River (USGS 09506000)

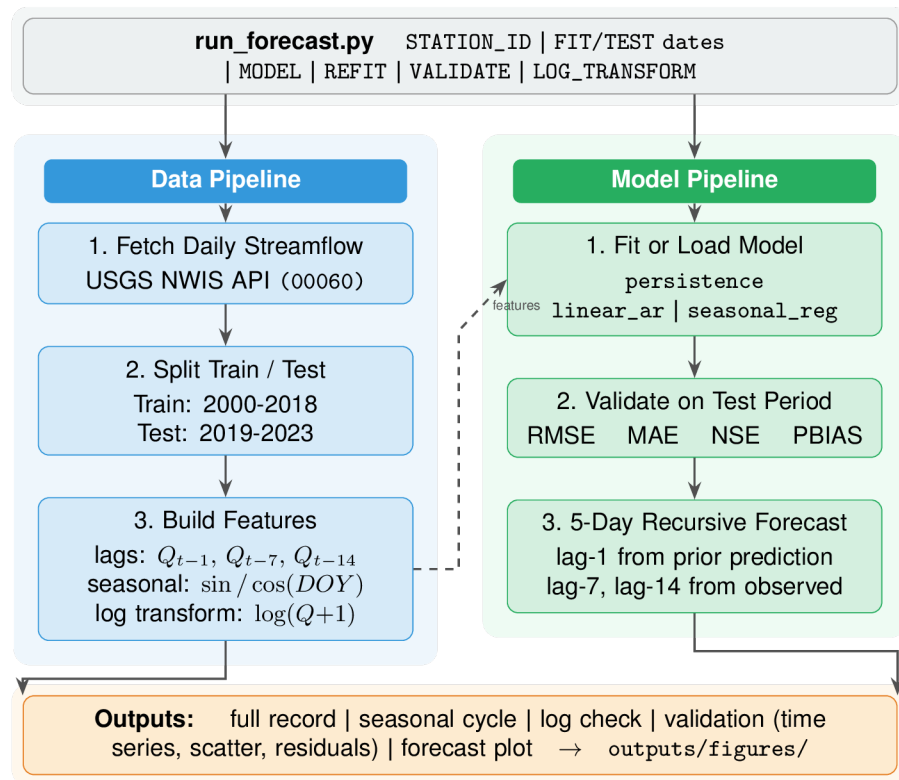


Figure 1: workflow diagram

Workflow Updates

I started from the class workflow (USGS.py) which just fetches data from the USGS API and makes a time series plot. Here is what I added:

1. **Model fitting and saving.** The original script stopped at downloading the data. I added three model options with a training and testing split. Fitted model coefficients get saved to a JSON file so you can reload without retraining.
2. **Log transform.** I fit the models on $\log(Q+1)$ instead of raw Q . Streamflow is right-skewed so working in log space helps with variance and gives better predictions at low flows. The back-transform happens automatically before any metric or plot.

3. **Recursive multi-step forecast.** The class script just fetched and plotted historical data. I added the actual forecasting logic where each predicted day feeds into the next step as the lag-1 feature.
4. **Validation metrics and plots.** Added RMSE, MAE, NSE, and percent bias on the held-out test period. The validation figure has three panels: time series with test predictions, a predicted vs observed scatter, and a residual histogram.
5. **Seasonal cycle and distribution plots.** Monthly mean with std bars, and a side-by-side histogram showing why the log transform makes sense.
6. **User config at top.** All the options (station, date ranges, forecast start, model choice, refit flag) are just variables at the top of the script.
7. **environment.yml** so the environment is reproducible.

New Model Added: Seasonal Regression

The two models from class were persistence ($Q_{\text{forecast}} = Q_{\text{yesterday}}$) and a simple linear autoregression with lag-1 streamflow.

The model I added is a seasonal multiple regression using: - Lagged streamflow at 1, 7, and 14 days back - sin and cos of day-of-year at the annual and semi-annual periods

So the equation is:

$$\begin{aligned} \log(Q+1)[t] = & a1*\log(Q+1)[t-1] + a2*\log(Q+1)[t-7] + a3*\log(Q+1)[t-14] \\ & + b1*\sin(2\pi*DOY/365) + b2*\cos(2\pi*DOY/365) \\ & + b3*\sin(4\pi*DOY/365) + b4*\cos(4\pi*DOY/365) \\ & + \text{intercept} \end{aligned}$$

The Fourier terms work because you can approximate any periodic signal with sin/cos pairs. The annual and semi-annual harmonics capture the winter baseflow, spring snowmelt peak, and monsoon pattern. This is different from lag-1 linear AR because it accounts for what time of year it is, not just what flow was yesterday. The 7-day and 14-day lags help with slower recession behavior after storms.

Still a linear regression fit with OLS, just with more features.

Note on AI

Used AI, still read and edited a lot of it myself.

Reflection Questions

Are you happy with how your repo came out?

For the most part yeah. The seasonal regression model works noticeably better than the simple AR which is good to see. The config block at the top is clean and the validation outputs are easy to read. If I was handing this to someone to run I would feel okay about it.

What was the most challenging aspect of this assignment and activities for you?

Getting the recursive forecasting indexing right was annoying. You have to track which lag values come from actual observed data and which come from your own previous predictions. For a 5-day forecast the 7 and 14-day lags are always from real data since you are only going 5 days out, but lag-1 flips to a model output starting on day 2. Getting that correct took a while. Also the log transform adds a layer where you have to be careful about when to transform and when to back-transform especially inside the recursive loop.

If you had more time, what would you have done differently?

I would have tried a machine learning model like a random forest or LSTM to see if it actually does better than the regression approach for this kind of forecast.

How do you think your skills have progressed this semester?

I came in with a decent background through my research so a lot of the Python side was not new to me. But this class still helped fill in gaps I did not know I had. The geospatial stuff especially, I had basically never worked with coordinate reference systems or rasters before HW5 and that was genuinely new. The workflow and reproducibility framing from this assignment is also something I had not thought about much before. I mostly just had scripts that ran on my machine and called it good.

Are you happy with this progress or are there things you wished we had done more or less of?

Yeah for the most part. I wish we had touched on some machine learning since that is where a lot of hydrology is going and I think it would have fit well with what we did. But overall the class covered good ground.